

Textual Analysis with Large Language Models (LLMs) –Tutorial

This document adds a background on the evolution of NLP to LLMs, extends the guidance for each step with incremental instructions, provides LLM API examples for several methods, and includes hypothetical model performance evaluation figures.

Evolution of NLP → LLMs (Definitions & Mechanisms)

Bag-of-Words (BoW)

In the earliest stage of NLP, text was represented simply as a “bag” of words, ignoring grammar and order.

- How it works: Count how many times each word appears in a document.
- Example: The sentence “Profits increased this quarter” → {profits:1, increased:1, this:1, quarter:1}.
- Limitation: The sentences “Profits increased this quarter” and “This quarter, profits increased” look identical to BoW.

TF-IDF (Term Frequency – Inverse Document Frequency)

This approach improves BoW by giving more weight to words that are frequent in one document but rare across many documents.

- How it works: Multiply the frequency of a word in a document by the inverse of how many documents contain that word.
- Example: In financial filings, the word “company” appears everywhere, so TF-IDF reduces its weight. The word “blockchain” may appear in only a few documents, so it gets higher importance.
- Limitation: Still treats text as unordered counts.

N-grams

Instead of single words, we look at fixed-length word sequences.

- How it works: Extract pairs (bigrams), triples (trigrams), etc.
- Example: The sentence “stock market crash” as bigrams becomes [stock market, market crash].
- Benefit: Captures short context like “interest rate” or “credit risk.”
- Limitation: Explodes in size if we try to capture longer sequences.

Part-of-Speech (POS) Tagging & Parsing

Rather than treating words as isolated tokens, POS tagging identifies their role in a sentence. Parsing then shows how words depend on each other.

- How it works: Assign tags such as noun (N), verb (V), adjective (ADJ). Dependency parsing links subjects and verbs.
- Example: In “The CEO announced strong earnings”: “CEO” → noun, “announced” → verb, “earnings” → noun (object).
- Benefit: Provides structure and grammar awareness.
- Limitation: Older parsers were brittle when facing slang or messy real-world text.

Word Embeddings (Word2Vec, GloVe)

A major leap came with word embeddings — representing each word as a dense vector in a multi-dimensional space.

- How it works: Train a neural network to predict a word from its context (or vice versa). Words used in similar contexts end up near each other in the vector space.
- Example: The relationship king – man + woman $\approx$ queen emerges.
- Benefit: Captures semantics: “stock” is closer to “equity” than to “banana.”
- Limitation: Each word has only one vector, so “bank” (finance vs. riverbank) gets confused.

Sequence Models – RNN, LSTM

These models process text word by word, remembering context with a hidden state.

- How it works: RNNs loop through tokens; LSTM add gates to better remember important words and forget irrelevant ones.
- Example: In “The central bank raised interest rates because inflation was high,” the model needs to remember “raised” until it reads “because inflation...”
- Benefit: Models order and context.
- Limitation: Struggles with very long sentences — information gets lost after many words.

Attention

Attention allows the model to focus selectively on certain words when interpreting another.

- How it works: Each word looks at every other word and decides how much to “pay attention” to it.

- Example: In “The bond market, which had been volatile, stabilized after the announcement,” the word “stabilized” pays attention to “bond market.”
- Benefit: Captures long-range dependencies much better than RNNs.

Transformers

The Transformer architecture (2017) put attention at its core. Instead of processing sequentially, it processes all tokens in parallel.

- How it works: Each token attends to all others at once, plus positional encodings tell the model about order.
- Example: Models like BERT (encoder-only) understand context, while GPT (decoder-only) generates text. T5 (encoder-decoder) handles translation.
- Benefit: Scales to huge datasets, powers today’s LLMs.

Pretraining Objectives

Transformers are trained on massive corpora using self-supervised objectives:

- MLM: Mask words and predict them (“The company [MASK] profits” → “reported”).
- CLM: Predict next word (“The stock price went” → “up”).
- Seq2Seq: Translate or summarize text.
- Benefit: Models learn language knowledge without labeled datasets.

Tokenization

Words are broken into smaller pieces (subwords) to handle rare or unknown terms.

- Example: “unhappiness” → [“un”, “happy”, “ness”].
- Benefit: Efficiently represents vocabulary.
- Relevance: Financial jargon like “fintechization” can still be split into meaningful parts.

Fine-tuning & Instruction Tuning

Pretrained models are adapted to specific tasks or made more user-friendly.

- Fine-tuning: Train further on domain-specific labeled data (e.g., FinBERT).
- Instruction Tuning: Train models to follow prompts like “Summarize this report in 3 bullets.”

RLHF (Reinforcement Learning with Human Feedback)

Aligns LLMs with human preferences.

- How it works: Humans rate outputs, a reward model is trained, then the LLM is optimized to maximize reward.
- Example: If the model answers “GDP is 5%” vs. “GDP is 5%, according to IMF 2023,” the latter is preferred. RLHF nudges toward better outputs.

Embeddings & Vector Search

LLMs can generate embeddings (numerical vectors) for chunks of text.

- Use case: Store filings as embeddings; retrieve similar passages by cosine similarity.
- Benefit: Powers Retrieval-Augmented Generation (RAG), semantic search, clustering.

Prompting

Prompts are the instructions we give to LLMs.

- Zero-shot: Ask directly.
- Few-shot: Provide examples first.
- Chain-of-thought: Ask to explain step-by-step.
- Example: Zero-shot: “Classify sentiment: ‘The firm expects growth.’” Few-shot: provide examples before asking.

Limitations & Risks

Even state-of-the-art LLMs have weaknesses.

- Hallucinations: Generate false facts.
- Context limits: Only remember fixed tokens.
- Biases: Reflect training data.
- Costs: Expensive.
- Mitigation: RAG (Retrieval-Augmented Generation), validation, grounding in real data.

B. Project Guidance (Extended, Step-by-Step)

Step 1 — Define a Research Question

- Economic hypothesis

Y variable: some economic outcomes (e.g., stock return)

X variable: features extracted from text data using LLM (e.g., market sentiment from news) – Information that cannot be extracted from numbers.

Step 2 — Collect Data

- Sources, metadata schema, storage layout, compliance and rate limits.

*** Following datasets are available upon request (contact me if you need them):**

1. Earnings call transcripts (global coverage)

gvkey	headline	chunk_id	chunk							
1045	AMR Corporation, Q4 2009 Earnings Call, Jan-20-2010	1	Thank you, Chris. Good afternoon, everyone. As you have seen in our press release excl							
1045	AMR Corporation, Q4 2009 Earnings Call, Jan-20-2010	2	What's more I believe we are in demonstratively better shape than we were a year ago.							
1045	AMR Corporation, Q4 2009 Earnings Call, Jan-20-2010	3	Our industry has certainly been tested with a variety of challenges over the past decade							
1045	AMR Corporation, Q4 2009 Earnings Call, Jan-20-2010	4	These changes will be finalized in April and will provide the foundation for our network							
1045	AMR Corporation, Q4 2009 Earnings Call, Jan-20-2010	5	We will be taking 45 Boeing 737-800 this year as part of our fleet renewal plan and we f							
1045	AMR Corporation, Q4 2010 Earnings Call, Jan 19, 2011	1	Okay, thank you, Chris. Good afternoon, everyone. As you have seen in our press releas							
1045	AMR Corporation, Q4 2010 Earnings Call, Jan 19, 2011	2	But first, let me just highlight a couple of things. In 2010, we took several major steps to							
1045	AMR Corporation, Q4 2010 Earnings Call, Jan 19, 2011	3	Last week we announced that we expect our joint business with Japan Airlines to launch							
1045	AMR Corporation, Q4 2010 Earnings Call, Jan 19, 2011	4	We are also very pleased to welcome our long-time partner Qantas to Dallas/Fort Wort							
1045	AMR Corporation, Q1 2011 Earnings Call, Apr 20, 2011	1	Thank you, Chris. Good afternoon, everyone. As you saw earlier today in our press relea							
1045	AMR Corporation, Q1 2011 Earnings Call, Apr 20, 2011	2	In fact during the first quarter, American initiated over a dozen of domestic fare increas							
1045	AMR Corporation, Q1 2011 Earnings Call, Apr 20, 2011	3	This cut is in addition to the capacity reduction we announced in March and further dem							
1045	AMR Corporation, Q1 2011 Earnings Call, Apr 20, 2011	4	And I think it's a great testament to the strength and quality of our partnership that we v							
1045	AMR Corporation, Q1 2011 Earnings Call, Apr 20, 2011	5	As you've seen, we recently filed an antitrust lawsuit against Travelport and Orbitz, since							
1045	AMR Corporation, Q1 2011 Earnings Call, Apr 20, 2011	7	Given the current oil trends and volatility, it's also important to mention that in March, v							
1045	AMR Corporation, Q1 2011 Earnings Call, Apr 20, 2011	8	In addition, we anticipate that within our current fleet, aircraft that are Section 1110-eli							
1045	AMR Corporation, Q2 2011 Earnings Call, Jul 20, 2011	1	Thank you, Chris. Good afternoon, everyone. As you've seen from our earlier announce							
1045	AMR Corporation, Q2 2011 Earnings Call, Jul 20, 2011	2	Over the last couple of years, weâ€™ve significantly strengthened our flexibility to make							
1045	AMR Corporation, Q2 2011 Earnings Call, Jul 20, 2011	3	There are numerous benefits to the fleet transactions we announced, including the fact							

Raw data from Seekingalpha:

<https://seekingalpha.com/article/4822287-rent-the-runway-inc-rent-q2-2025-earnings-call-prepared-remarks-transcript>

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Explore Investing Groups

PORTFOLIOS

Portfolio

Portfolio Health Check

Create Portfolio +

FIND & COMPARE

Top Stocks

Top ETFs

Stock Screener

ETF Screener

Symbols, Analysts, Keywords

Today at 12 PM ET: 6 Stocks Poised to Benefit

Transcripts

Rent The Runway, Inc. (RENT) Q2 2025 Earnings Call Transcript

Sep. 15, 2025 9:15 AM ET | Rent the Runway, Inc. (RENT) Stock | RENT

 SA Transcripts
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 Play Earnings Call

Rent the Runway, Inc. (NASDAQ:[RENT](#)) Q2 2025 Earnings Call
September 11, 2025 4:30 PM EDT

Company Participants

Cara Schembri - Chief Legal & Administrative Officer
Jennifer Hyman - Co-Founder, CEO, President & Chairman
Siddharth Thacker - Chief Financial Officer

Presentation

Operator

Rent The Runway, Inc. (RENT) Q2 2025 Earnings Call Transcript

Sep. 15, 2025 9:15 AM ET | **Rent the Runway, Inc. (RENT) Stock** | RENT

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Rent the Runway, Inc. (NASDAQ:[RENT](#)) Q2 2025 Earnings Call
September 11, 2025 4:30 PM EDT

Presentation

2. US firms' patent descriptions

A	B	C
app_pub_dates	gvkey	titles
20011227	6066	Driver circuit, receiver circuit, and semiconductor integrated circuit device
20010816	4510	Floating spar for supporting production risers
20010712	10016	Device for sensing and locating objects hidden behind accessible surfaces of architectural structures
20010705	2136	Automatic called party locator over internet
20010628	14145	Method and apparatus for initiating telephone calls using a data network
20010531	62599	Optical node system for a ring architecture and method thereof
20011004	23197	Data flow control unit
20010927	21139	Loudspeaker system
20010927	63804	Chimeric proteins for use in transport of a selected substance into cells
20010823	1704	Consecutive deposition system
20011213	6109	Circuit for synchronous rectification with minimal reverse recovery losses
20010920	133729	Surgical instrument with off-axis electrode
20010524	5439	Underbalanced well completion
20010823	6066	Process for forming cone shaped solder for chip interconnection
20010517	9545	Energy absorbing disc travel limiter
20010830	24040	Generation of virtual combinatorial libraries of compounds
20010712	6066	Enqueueing apparatus for asynchronous transfer mode (ATM) virtual circuit merging
20010712	6066	Interleaved processing system for processing frames within a network router
20010823	6066	Method for transmitting loopback cells through a switching node of an asynchronous transfer mode (ATM) network
20010816	5047	Debris trap in a turbine cooling system



US 20080010203A1

(19) **United States**

(12) **Patent Application Publication Grant** (10) **Pub. No.: US 2008/0010203 A1**
(43) **Pub. Date: Jan. 10, 2008**

(54) **PURCHASING ALERT METHODS AND APPARATUS**

Related U.S. Application Data

(60) Provisional application No. 60/609,591, filed on Sep. 13, 2004.

(76) Inventor: **David S. Grant**, Mission Viejo, CA (US)

Publication Classification

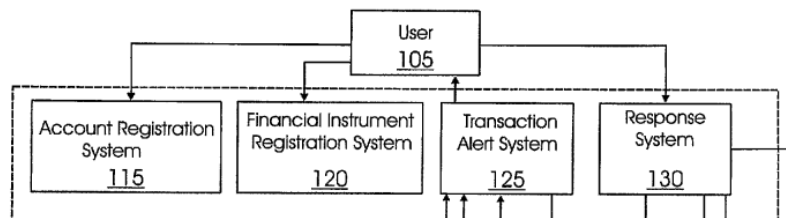
Correspondence Address:
MACPHERSON KWOK CHEN & HEID LLP
2033 GATEWAY PLACE
SUITE 400
SAN JOSE, CA 95110 (US)

(51) **Int. Cl.**
G06Q 40/00 (2006.01)
(52) **U.S. Cl.** **705/44**

(57) **ABSTRACT**

Systems and techniques for receiving transaction information at an authentication system. The transaction information may be processed using different rule sets, based on the financial instrument used for the transaction (for example, a particular credit card, debit card, bank account, brokerage account, and the like). One or more alerts may be generated and communicated to the user. The alerts may be formatted based on user device configuration information, such as a cell phone type, email type, and the like.

(21) Appl. No.: **11/572,413**
(22) PCT Filed: **Sep. 9, 2005**
(86) PCT No.: **PCT/US05/32068**
§ 371(c)(1),
(2), (4) Date: **Jan. 19, 2007**



4. Official task descriptions required by each job occupation



O*NET® 30.0 Database

The O*NET database contains a rich set of variables that describe work and worker characteristics, including skill requirements. Developers and other customers are encouraged to incorporate the O*NET database within their products, services, and research. This section contains an overview of the most recent database and a variety of database download options, as well as format-specific data dictionaries.

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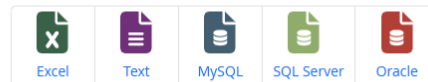
The O*NET 30.0 Database is available under a Creative Commons license. See the [Database license page](#) for more information.

- [Knowledge, Skills, Abilities](#)
- [Work Values and Work Styles](#)
- [Work Activities](#)
- [Related Occupations and Related Domains](#)
- [Education, Experience, Training](#)
- [Tasks](#)
- [Work Context](#)
- [Data Collection](#)
- [Interests](#)
- [Technology Skills & Tools](#)
- [Occupation Titles](#)
- [Competency Frameworks](#)

Knowledge, Skills, Abilities

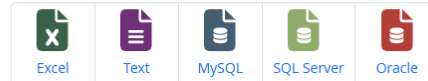
Knowledge updated in 30.0

This file contains the Content Model Knowledge data associated with each O*NET-SOC occupation. [Read more...](#)



Skills updated in 30.0

This file contains the Content Model Skill data associated with each O*NET-SOC occupation. [Read more...](#)



5. US firms' annual report (https://sraf.nd.edu/sec-edgar-data/lm_10x_summaries/)

Software Repository for Accounting and Finance

[Home](#) [SEC/EDGAR Data](#)

10X Summaries

Loughran-McDonald_10X_Summaries_1993-2024.csv (188MB)

Using the [cleaned files from the stage one parsing process](#), a dataset is created containing summary data for each filing. A [Python class for this module is available](#). The file contains a header record with labels and is comma-delimited. Each record reports:

Home

Loughran-McDonald Master Dictionary w/ Sentiment Word Lists

Complexity

SEC/EDGAR Data

Merger-related filings


```

K-TRON
INTERNATIONAL, INC.
(Name of Registrant as Specified in
Its Charter)
N/A
(Name of Person(s) Filing Proxy
Statement, if other than Registrant)
Payment of Filing Fee (Check the appropriate box):
Routes 55
& 553
P.O. Box 888
Pitman, New Jersey
08071-0888
(856) 589-0500
February 26,
2010
To our Shareholders:
You are cordially invited to attend a special meeting of the
shareholders of K-Tron International, Inc. (the
[Company]), at the offices of Morgan,
Lewis & Bockius LLP located at 1701 Market Street,
Philadelphia, Pennsylvania 19103 on Thursday, April 1,
2010, beginning at 10:00 a.m. local time.
At the special meeting, you will be asked to consider and vote
upon a proposal to approve the Agreement and Plan of Merger
([Merger Agreement]), dated as of January 8,
2010, among Hillenbrand, Inc. ([Parent]), Krusher
Acquisition Corp., a wholly-owned subsidiary of Parent
([Merger Sub]), and the Company.
The Merger Agreement provides for, among other things, the
merger of Merger Sub with and into the Company, with the Company
as the surviving corporation in the merger (the

```

Ln 102, Col 37 | 531,121 characters | Plain text

- Other data from website through web scrawling:

Sample code:

```

import time, random, requests
from bs4 import BeautifulSoup

```

```

HEADERS = {"User-Agent": "Mozilla/5.0 (compatible; EduCrawler/1.0;
+https://example.edu/crawler-info)"}

```

```

def fetch(url, sleep=(1,3)):
    time.sleep(random.uniform(*sleep))
    r = requests.get(url, headers=HEADERS, timeout=15)
    r.raise_for_status()
    return r.text

```

```

html = fetch("https://example.com/")
soup = BeautifulSoup(html, "html.parser")
print(soup.title.get_text(strip=True))

```

Step 3 — Preprocess Text

- Cleaning, segmentation/chunking with overlap

Step 4 — Choose Method(s) (LLM API Examples Included)

4A) Sentiment (API):

```
import requests, json
endpoint = "https://api.example-llm.com/v1/chat/completions"
headers = {"Authorization": "Bearer YOUR_API_KEY", "Content-Type":
"application/json"}
prompt = "Classify sentiment (Positive/Neutral/Negative): 'We anticipate margin
expansion next quarter.'"
payload = {"model": "your-sentiment-
model", "messages": [{"role": "user", "content": prompt}], "temperature": 0}
res = requests.post(endpoint, headers=headers, data=json.dumps(payload)).json()
print(res["choices"][0]["message"]["content"])
```

4B) Zero-shot labels (API):

```
labels = ["Innovation", "Risk", "Guidance", "ESG"]
prompt = f'Pick ONE label from {labels} for: 'R&D investments in AI have accelerated.'
Return ONLY the label.'
payload["messages"] = [{"role": "user", "content": prompt}]
```

4C) Summarization (API):

```
long_text = "..." # your document chunk
prompt = "Summarize in 3-4 bullets focusing on forward-looking statements:\n" +
long_text
payload["messages"] = [{"role": "user", "content": prompt}]
```

4D) Embeddings (API):

```
emb_endpoint = "https://api.example-llm.com/v1/embeddings"
E = requests.post(emb_endpoint, headers=headers, data=json.dumps({"model": "your-
embedding-model", "input": ["A", "B"]})).json()["data"]
vecA, vecB = E[0]["embedding"], E[1]["embedding"]
```

4E) Information Extraction (API JSON schema):

```
schema = '{"growth_rate":"number","risk_flags":"string[]}'
text = "We expect 5% revenue growth; risks include regulation and FX."
prompt = f'Extract fields and return ONLY valid minified JSON matching
{schema}:\n{text}'
payload["messages"] = [{"role":"user","content":prompt}]
```

4F) RAG (retrieve, then ask LLM):

```
from sklearn.neighbors import NearestNeighbors
import numpy as np
vectors = np.random.randn(100, 768) # placeholder embeddings
nbrs = NearestNeighbors(n_neighbors=5, metric="cosine").fit(vectors)
q_vec = np.random.randn(1, 768)    # embedding(query)
dist, idx = nbrs.kneighbors(q_vec)
context = "\n".join([f"[Doc {i}] ..." for i in idx[0]])
prompt = f'Use ONLY this context to answer. If insufficient, say
so.\nContext:\n{context}\n\nQuestion: ...'
```

Step 5 — Construct Features

- Aggregate per firm–time: tone, tag frequencies, novelty; align to structured controls; record provenance.

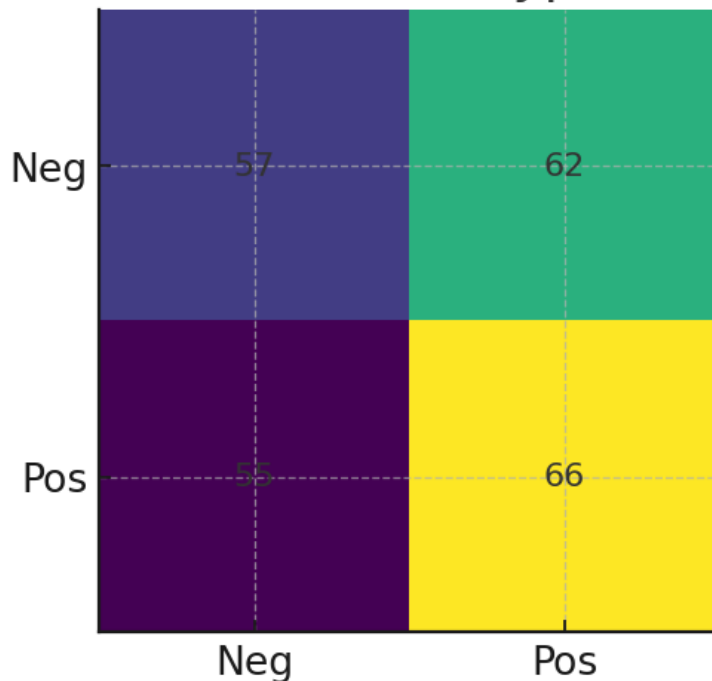
Step 6 — Empirical Testing

- Predictive and economic tests; robustness checks; ablations (dictionary vs. LLM).

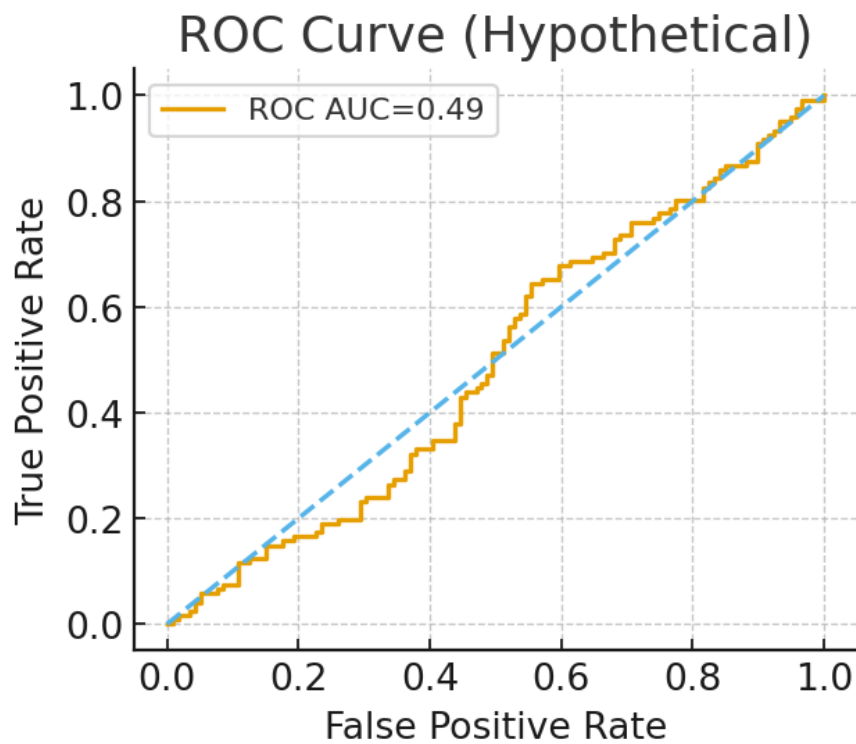
Step 7 — Evaluation & Visualization (Examples)

Confusion Matrix:

Confusion Matrix (Hypothetical)

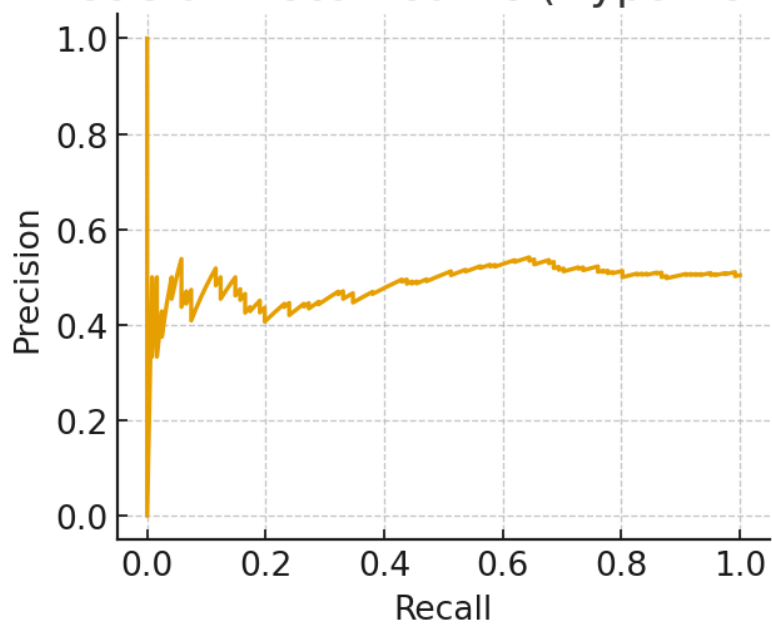


ROC Curve:

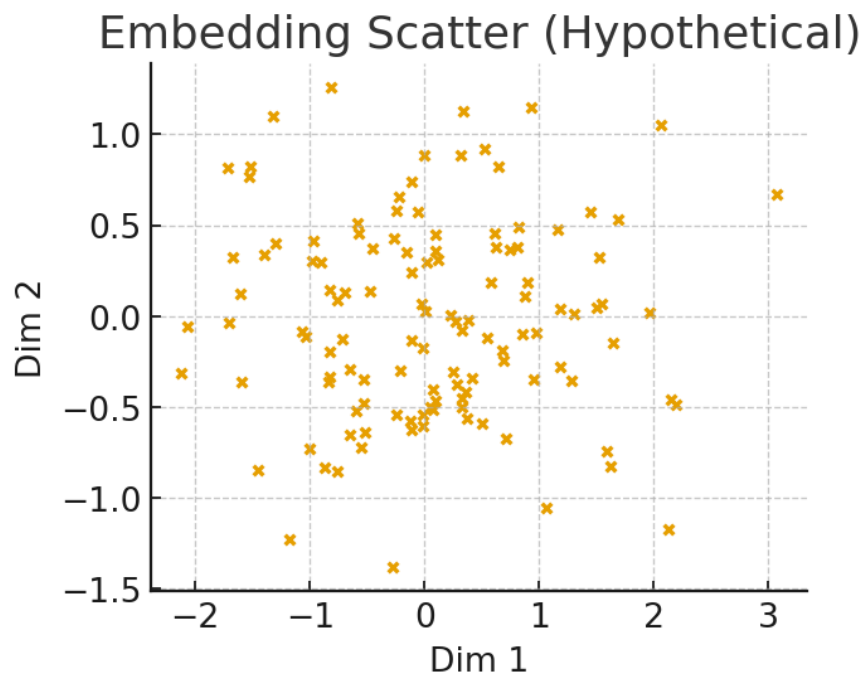


Precision-Recall Curve:

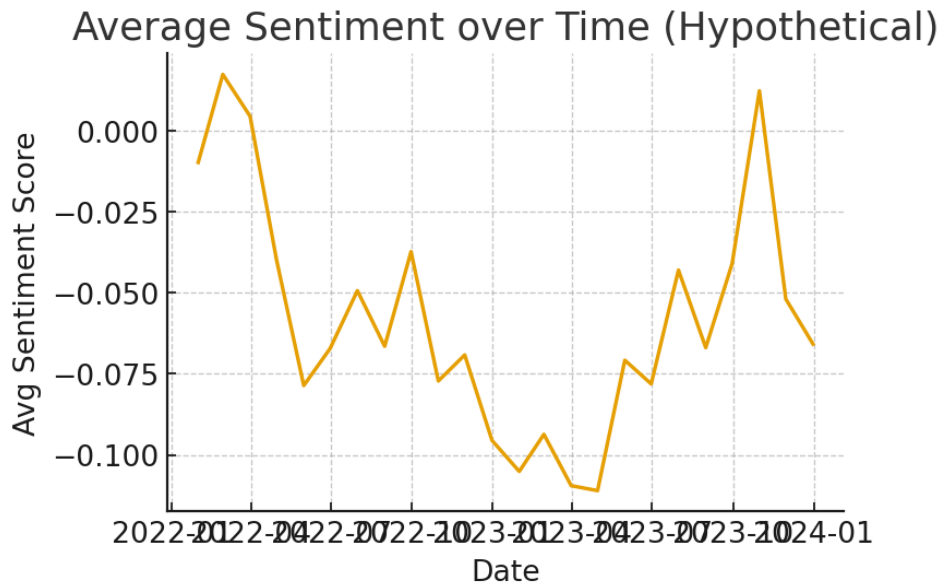
Precision-Recall Curve (Hypothetical)



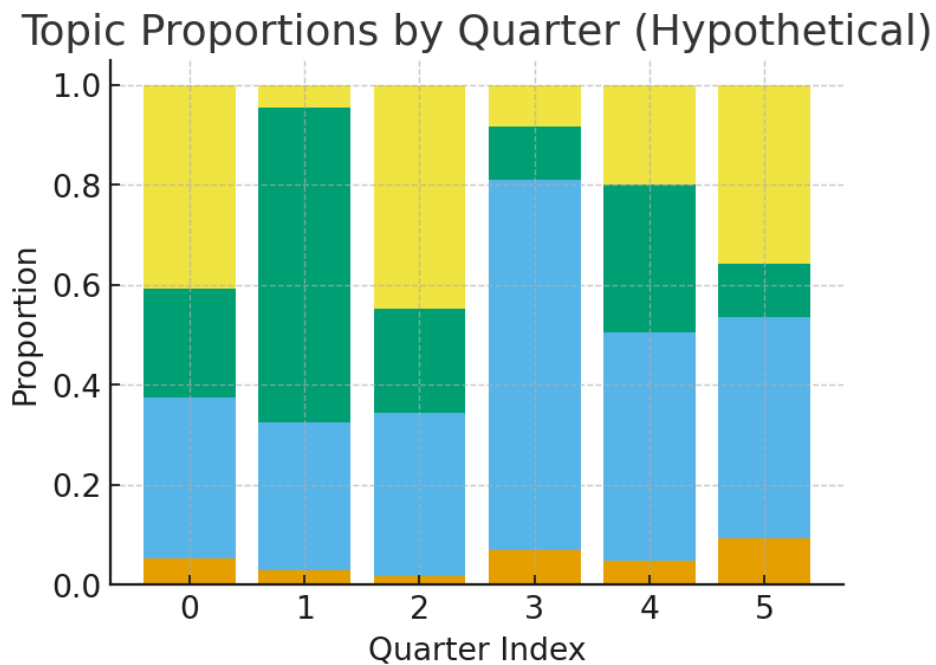
Embedding Scatter:



Average Sentiment over Time:



Topic Proportions:



Step 8 — Interpretation & Finance Theory

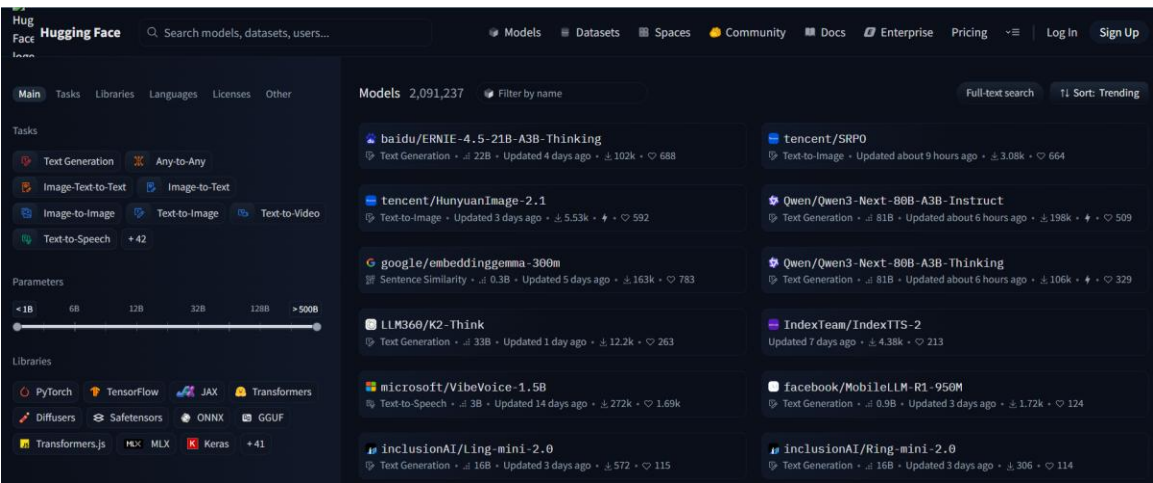
- Link findings to market efficiency, information asymmetry, and behavioral channels; discuss external validity.

Step 9 — Report Writing

- Include references/citations, model/prompt details, results, summary statistics, statistical + economic significance, and limitations.

Resources & Platforms

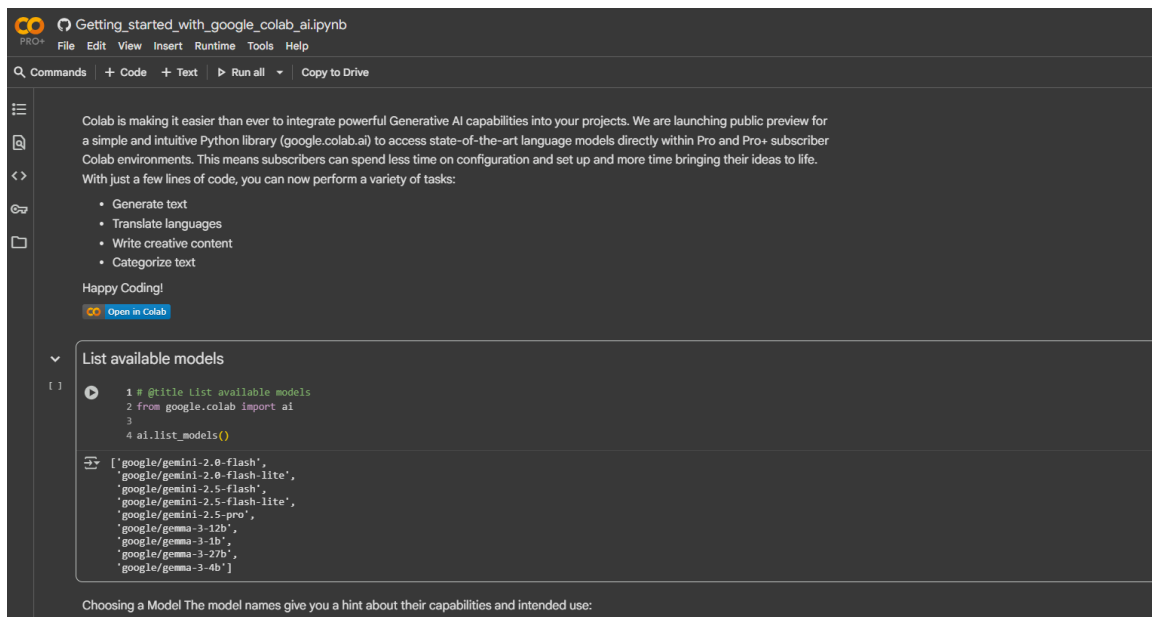
<https://huggingface.co/models>



<https://dmxapi.cn/models.html#code-block>



[https://colab.research.google.com/github/googlecolab/colabtools/blob/main/notebooks/Getting started with google colab ai.ipynb#scrollTo=Ucchuu5vV3jp](https://colab.research.google.com/github/googlecolab/colabtools/blob/main/notebooks/Getting%20started%20with%20google%20colab%20ai.ipynb#scrollTo=Ucchuu5vV3jp)



```
Getting_started_with_google_colab_ai.ipynb
File Edit View Insert Runtime Tools Help

Commands + Code + Text ▶ Run all ▾ Copy to Drive

Colab is making it easier than ever to integrate powerful Generative AI capabilities into your projects. We are launching public preview for a simple and intuitive Python library (google.colab.ai) to access state-of-the-art language models directly within Pro and Pro+ subscriber Colab environments. This means subscribers can spend less time on configuration and set up and more time bringing their ideas to life.

With just a few lines of code, you can now perform a variety of tasks:

• Generate text
• Translate languages
• Write creative content
• Categorize text

Happy Coding!
Open in Colab

List available models

1 # @title List available models
2 from google.colab import ai
3
4 ai.list_models()

['google/gemini-2.0-flash',
 'google/gemini-2.0-flash-lite',
 'google/gemini-2.5-flash',
 'google/gemini-2.5-flash-lite',
 'google/gemini-2.5-pro',
 'google/gemma-3-12b',
 'google/gemma-3-1b',
 'google/gemma-3-27b',
 'google/gemma-3-4b']

Choosing a Model The model names give you a hint about their capabilities and intended use:
```

References:

Lopez-Lira, A., & Tang, Y. (2023). Can chatgpt forecast stock price movements? return predictability and large language models. *arXiv preprint arXiv:2304.07619*.

Chen, M. A., & Wang, J. (2025). Displacement or augmentation? The effects of AI innovation on workforce dynamics and firm value. *SSRN*.